Matrices, vectors, and cases

In `mat`, a semicolon ends a row and a comma separates columns:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

`vec` stacks its arguments into a column vector:

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Pick the fence with `delim`. Square brackets, or bars for a determinant:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

`cases` branches a definition; the & lines the conditions up:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$